

EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB REPORT # 3



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.
Kaynat Rana

Experiment #3

Implementation on singly Linked List

Objective:

The objective of this session is to understand the various operations on Singly linked list in C++.

1. Insertion
2. Deletion
3. Traversal

Equipment Used:

Visual studio C++

Procedure:

Singly Linked List:

A singly linked list is a type of linked list that is unidirectional, that is it can be traversed in only one direction from the head to the last node. Each element in a linked list is called a node. A single node contains data and a pointer to the next node, which helps to maintain the structure of the list.

Linked List:

A linked list is a linear dynamic data structure. The node of a singly or a circular linked list has two parts: data and address of the next node. The node of a double linked list has three parts: data, the address of the previous node and the address of the next node.

Syntax/general form:

```
Struct node
{
int data;
node*next;
};
```

The pointer variable declaration is as follows:

node*head;

Operation on Linked List:

Following operations can be performed on a linked list:-

1.Traversal:

To traverse all the nodes one after another.

2.Insertion:

To add a node at the given position.

3.Deletion:

To delete a node.

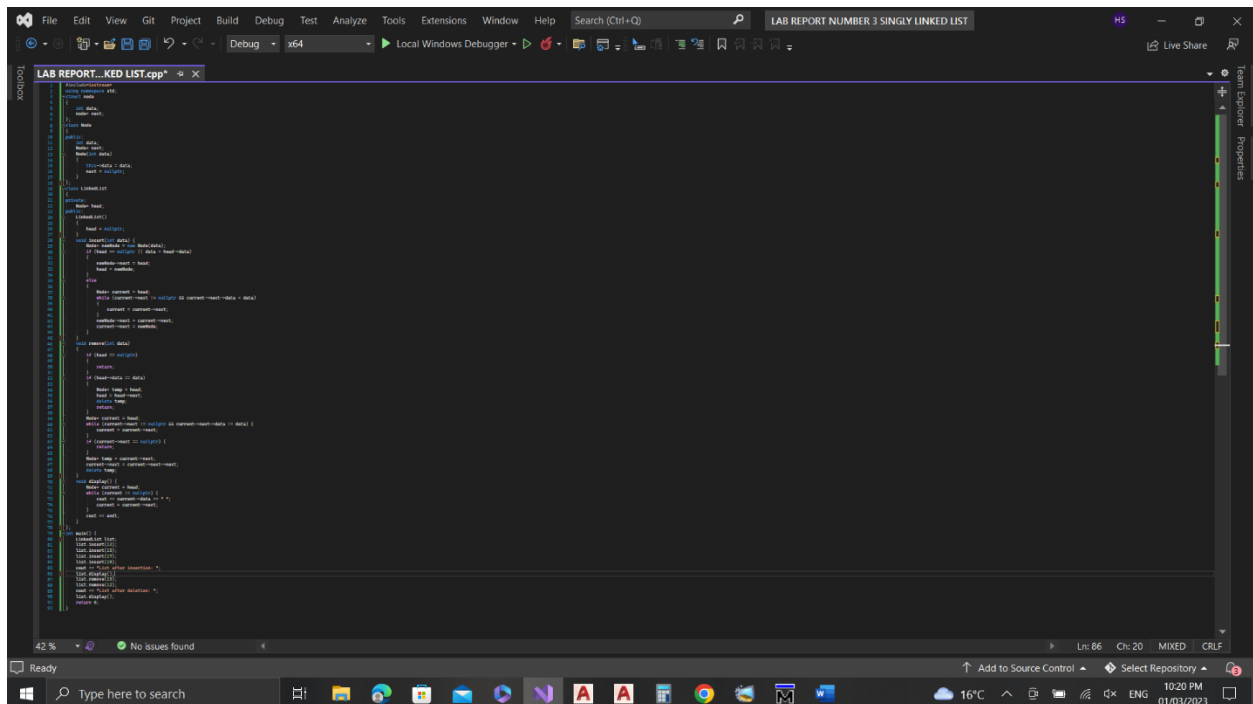
Tasks

Task 1:

1. Write a C++ program to create Singly Linked List ADT. Your Linked List must be able to perform the following functionalities:

- (a) Insertion of an integer (start/middle/end) in sorted linked list.
- (b) Deletion of a given integer from the above linked list.
- (c) Display the contents of the above list after deletion.

CODE:



```
LAB REPORT_KED LIST.cpp
// Singly Linked List ADT
// Insertion, Deletion, Display
// Author: [Your Name]

#include <iostream>
using namespace std;

// Node structure
struct Node {
    int data;
    Node* next;
};

// Singly Linked List structure
struct SinglyLinkedList {
    Node* head;
    Node* tail;
};

// Function to create a new node
Node* createNode(int data) {
    Node* newNode = new Node;
    newNode->data = data;
    newNode->next = nullptr;
    return newNode;
}

// Function to insert a new node at the end of the list
void insertEnd(SinglyLinkedList* list, int data) {
    Node* newNode = createNode(data);
    if (list->head == nullptr) {
        list->head = newNode;
        list->tail = newNode;
    } else {
        list->tail->next = newNode;
        list->tail = newNode;
    }
}

// Function to delete a node from the list
void deleteNode(SinglyLinkedList* list, int data) {
    if (list->head == nullptr) return;

    if (list->head->data == data) {
        list->head = list->head->next;
    } else {
        Node* temp = list->head;
        while (temp->next != nullptr) {
            if (temp->next->data == data) {
                temp->next = temp->next->next;
            } else {
                temp = temp->next;
            }
        }
    }
}

// Function to display the contents of the list
void displayList(SinglyLinkedList* list) {
    if (list->head == nullptr) {
        cout << "List is empty" << endl;
    } else {
        Node* temp = list->head;
        while (temp != nullptr) {
            cout << temp->data << " ";
            temp = temp->next;
        }
        cout << endl;
    }
}

// Main function
int main() {
    SinglyLinkedList list;

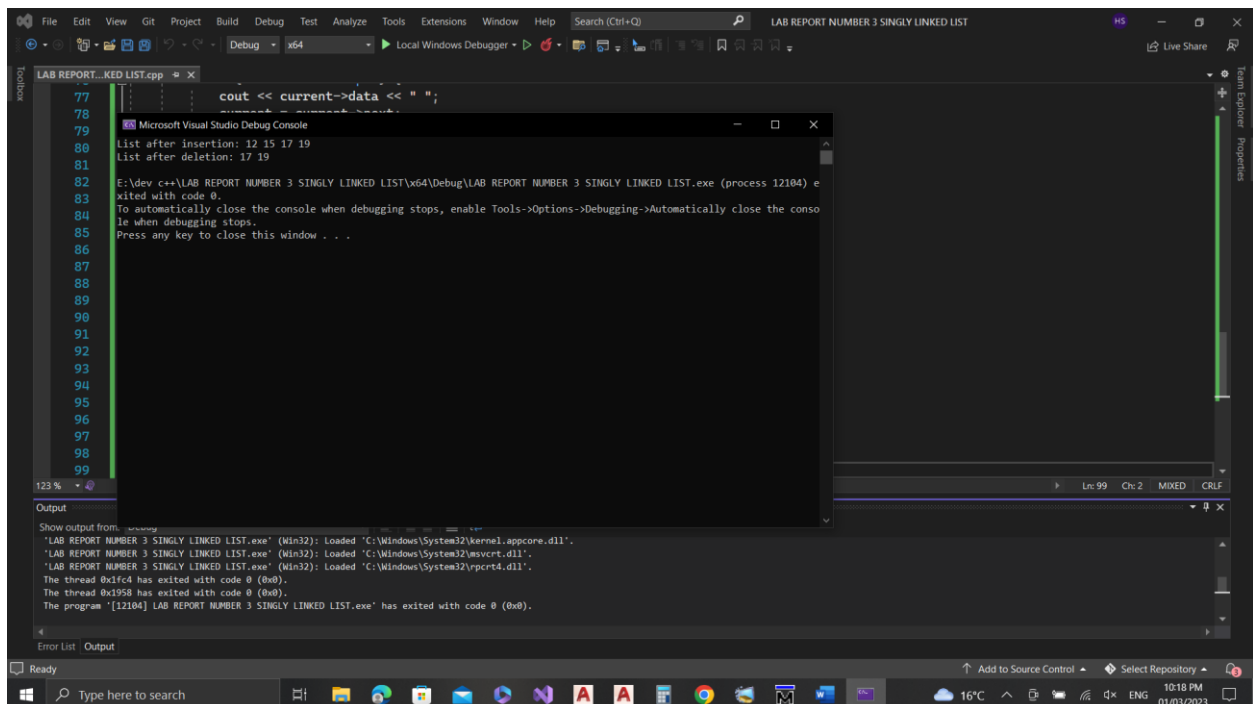
    // Insertion
    insertEnd(&list, 12);
    insertEnd(&list, 15);
    insertEnd(&list, 17);
    insertEnd(&list, 19);

    // Deletion
    deleteNode(&list, 15);

    // Display
    displayList(&list);

    return 0;
}
```

OUTPUT:



```
LAB REPORT_KED LIST.cpp
77 // Singly Linked List ADT
78 // Insertion, Deletion, Display
79 // Author: [Your Name]
80
81 #include <iostream>
82 using namespace std;
83
84 // Node structure
85 struct Node {
86     int data;
87     Node* next;
88 };
89
90 // Singly Linked List structure
91 struct SinglyLinkedList {
92     Node* head;
93     Node* tail;
94 };
95
96 // Function to create a new node
97 Node* createNode(int data) {
98     Node* newNode = new Node;
99     newNode->data = data;
100     newNode->next = nullptr;
101     return newNode;
102 }
103
104 // Function to insert a new node at the end of the list
105 void insertEnd(SinglyLinkedList* list, int data) {
106     Node* newNode = createNode(data);
107     if (list->head == nullptr) {
108         list->head = newNode;
109         list->tail = newNode;
110     } else {
111         list->tail->next = newNode;
112         list->tail = newNode;
113     }
114 }
115
116 // Function to delete a node from the list
117 void deleteNode(SinglyLinkedList* list, int data) {
118     if (list->head == nullptr) return;
119
120     if (list->head->data == data) {
121         list->head = list->head->next;
122     } else {
123         Node* temp = list->head;
124         while (temp->next != nullptr) {
125             if (temp->next->data == data) {
126                 temp->next = temp->next->next;
127             } else {
128                 temp = temp->next;
129             }
130         }
131     }
132 }
133
134 // Function to display the contents of the list
135 void displayList(SinglyLinkedList* list) {
136     if (list->head == nullptr) {
137         cout << "List is empty" << endl;
138     } else {
139         Node* temp = list->head;
140         while (temp != nullptr) {
141             cout << temp->data << " ";
142             temp = temp->next;
143         }
144         cout << endl;
145     }
146 }
147
148 // Main function
149 int main() {
150     SinglyLinkedList list;
151
152     // Insertion
153     insertEnd(&list, 12);
154     insertEnd(&list, 15);
155     insertEnd(&list, 17);
156     insertEnd(&list, 19);
157
158     // Deletion
159     deleteNode(&list, 15);
160
161     // Display
162     displayList(&list);
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164     return 0;
165 }
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A singly linked list is a data structure that consists of a sequence of nodes, where each node contains a data element and a pointer to the next node in the list. The first node in the list is called the head and the last node is called the tail.

